

Force Vectors & Resultants (review)

$F_R = \sqrt{F_1^2 + F_2^2 + 2F_1F_2\cos\theta}$ – parallelogram law, resultant magnitude (θ between F_1, F_2); $\frac{F_1}{\sin\theta_1} = \frac{F_2}{\sin\theta_2} = \frac{F_R}{\sin\theta_R}$ – law of sines gives its direction

$F_x = F\cos\theta$, $F_y = F\sin\theta$ – rectangular components (2D), θ from the x -axis

$F = \sqrt{F_x^2 + F_y^2}$, $\theta = \tan^{-1}\left(\frac{F_y}{F_x}\right)$ – magnitude/direction from components

$\mathbf{F} = F_x\mathbf{i} + F_y\mathbf{j} + F_z\mathbf{k}$, $F = \sqrt{F_x^2 + F_y^2 + F_z^2}$ – Cartesian vector notation (3D)

$\mathbf{u} = \frac{\mathbf{F}}{F} = \cos\alpha\mathbf{i} + \cos\beta\mathbf{j} + \cos\gamma\mathbf{k}$ – unit vector & direction cosines (α, β, γ from $+x, +y, +z$)

$\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$ – direction cosine identity

$\mathbf{r} = (x_2 - x_1)\mathbf{i} + (y_2 - y_1)\mathbf{j} + (z_2 - z_1)\mathbf{k}$ – position vector, tail to head

$\mathbf{u} = \frac{\mathbf{r}}{r}$, $\mathbf{F} = F\mathbf{u}$ – force along a line

$\mathbf{F}_R = \sum \mathbf{F} = (\sum F_x)\mathbf{i} + (\sum F_y)\mathbf{j} + (\sum F_z)\mathbf{k}$ – resultant of a concurrent force system: sum components independently

$\mathbf{A} \cdot \mathbf{B} = A_xB_x + A_yB_y + A_zB_z = AB\cos\theta$ – dot product, θ between $\mathbf{A}, \mathbf{B}$; $\theta = \cos^{-1}\left(\frac{\mathbf{A} \cdot \mathbf{B}}{AB}\right)$; projection of $\mathbf{A}$ onto a line with unit vector $\mathbf{u}$ is $\mathbf{A} \cdot \mathbf{u}$

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

– cross product; $|\mathbf{A} \times \mathbf{B}| = AB\sin\theta$, direction by right-hand rule. Used for $\mathbf{M} = \mathbf{r} \times \mathbf{F}$

Moments, Couples & Force-Couple Systems

$M_O = Fd$ – scalar moment, d = perpendicular distance from O to the line of action of F ; sign by right-hand rule (CCW + common)

$$\mathbf{M}_O = \mathbf{r} \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ r_x & r_y & r_z \\ F_x & F_y & F_z \end{vmatrix}$$

– vector moment; $\mathbf{r}$ = position vector from O to any point on the line of action of $\mathbf{F}$ (preferred for 3D)

$\mathbf{M}_O = \sum(\mathbf{r} \times \mathbf{F}) = \sum(xF_y - yF_x)$ – Varignon's theorem: moment of a force = sum of moments of its components

$$M_a = \mathbf{u}_a \cdot (\mathbf{r} \times \mathbf{F}) = \begin{vmatrix} u_x & u_y & u_z \\ r_x & r_y & r_z \\ F_x & F_y & F_z \end{vmatrix}$$

– moment about a specified axis (triple scalar product); $\mathbf{u}_a$ along the axis, $\mathbf{r}$ from any point on the axis to any point on the line of action

$M = Fd$, $\mathbf{M} = \mathbf{r} \times \mathbf{F}$ – moment of a couple; couple = two equal, opposite, non-collinear forces; its moment is the same about every point (free vector)

$\mathbf{F}_R = \sum \mathbf{F}$, $\mathbf{M}_{R_O} = \sum \mathbf{M}_O + \sum(\mathbf{r} \times \mathbf{F})$ – equivalent force-couple system at a chosen point O

$d = \frac{M_{R_O}}{F_R}$ – reduces to a single resultant force (no couple) only when $\mathbf{F}_R \perp \mathbf{M}_{R_O}$; line of action shifts by d from O

Equilibrium of Rigid Bodies & Dry Friction

$\sum F_x = 0$, $\sum F_y = 0$, $\sum M_O = 0$ – equilibrium, 2D (3 independent scalar eqns; alternate sets like $\sum M_A = \sum M_B = \sum M_C = 0$,

A, B, C non-collinear, also valid)

$\sum F_x = 0$, $\sum F_y = 0$, $\sum F_z = 0$, $\sum M_x = 0$, $\sum M_y = 0$, $\sum M_z = 0$ – equilibrium, 3D (6 independent scalar eqns; equivalently $\sum \mathbf{F} = 0$, $\sum \mathbf{M}_O = 0$)

Two-force member: loaded at only two points, no applied couples $\Rightarrow$ forces at both points equal, opposite, and collinear with the line joining them

Three-force member: equilibrium under exactly three forces requires them to be concurrent or all mutually parallel

2D support reactions: roller/rocker: 1 unknown, $\perp$ to the surface – pin/hinge: 2 unknowns (R_x, R_y), no moment – fixed support: 3 unknowns (R_x, R_y, M), no translation or rotation – cable/two-force link: 1 unknown, tension along the member – smooth surface contact: 1 unknown, normal force

3D support reactions: ball-and-socket: 3 unknowns (F_x, F_y, F_z), no moment – single journal bearing: 4 unknowns (2 $\perp$ forces + 2 $\perp$ moments) – single thrust bearing: 5 unknowns (2 $\perp$ forces + axial force + 2 $\perp$ moments) – fixed support: 6 unknowns ($F_x, F_y, F_z, M_x, M_y, M_z$) – smooth surface/roller: 1 unknown, normal force

$F \leq \mu_s N$ (no slipping), $F = \mu_k N$ (slipping) – dry (Coulomb) friction; μ_s, μ_k = static/kinetic coefficients ($\mu_k < \mu_s$ typically), N = normal force

$\phi_s = \tan^{-1}(\mu_s)$ – angle of friction; at impending slip the resultant contact force makes angle ϕ_s with the normal. Always check both slipping and tipping

Distributed Forces – Centroids & Loads

$$\bar{x} = \frac{\int x dA}{\int dA}, \quad \bar{y} = \frac{\int y dA}{\int dA}$$

– centroid of an area: average position of the dA elements

$$\bar{x} = \frac{\int x dL}{\int dL}, \quad \bar{y} = \frac{\int y dL}{\int dL} \text{ – centroid of a line (wire/rod/curve)}$$

$$\bar{x} = \frac{\sum \bar{x}_i A_i}{\sum A_i}, \quad \bar{y} = \frac{\sum \bar{y}_i A_i}{\sum A_i} \text{ – composite bodies: locate each part's centroid, take the weighted average; subtract } A_i \text{ for holes}$$

$A = \theta \bar{r} L$, $V = \theta \bar{r} A$ – Pappus-Guldinus: surface area/volume of revolution; θ = angle of revolution (2π full turn), $\bar{r}$ = centroid distance from the axis

$F_R = \int w(x) dx$, $\bar{x} = \frac{\int x w(x) dx}{\int w(x) dx}$ – resultant of a distributed load on a beam: F_R = area under the load diagram, acts through its centroid

Centroids – common shapes: $\bar{x} = b/2$, $\bar{y} = h/2$ – rectangle ($b \times h$), geometric center – $\bar{y} = h/3$ – triangle centroid height, from the base – $\bar{x} = \bar{y} = 0$ – circle, at center – $\bar{y} = 4r/3\pi$ – semicircular area, from the diameter – $\bar{x} = \bar{y} = 4r/3\pi$ – quarter-circular area, from each straight edge

Centroids – lines & arcs: midpoint – straight line segment – $\bar{x} = \bar{y} = 2r/\pi$ – quarter-circular arc, from the center – $\bar{y} = 2r/\pi$ – semicircular arc, from the diameter

Trusses, Frames & Machines

$\sum F_x = 0$, $\sum F_y = 0$ (at each joint) – method of joints: isolate each pin as a particle; start where at most 2 unknowns act, work outward

$\sum F_x = 0$, $\sum F_y = 0$, $\sum M = 0$ (on the cut section) – method of sections: cut up to 3 unknown members, keep one side, solve directly for a specific member force

Zero-force members: two non-collinear members at a joint with no load/support there $\Rightarrow$ both are zero-force – three members at a joint

(two collinear), no external load there $\Rightarrow$ the third (non-collinear) member is zero-force

$m + r = 2j$ – static determinacy of a plane truss; m = members, r = reaction components, j = joints. =: determinate, <: unstable, >: indeterminate

Frames/machines (multi-force members): dismember at the pins; draw a separate FBD for each member loaded at 3+ points; apply $\sum F_x = \sum F_y = \sum M = 0$ to each, using Newton's 3rd law at shared pins

Internal Forces in Members

$\sum F_x = 0$, $\sum F_y = 0$, $\sum M = 0$ – internal loadings by the method of sections: cut at the section of interest, apply equilibrium to one segment for N, V, M

Sign convention: N positive in tension; V positive when it tends to rotate the segment clockwise; M positive when it bends the member concave up (sagging) – fix this before drawing N, V, M diagrams

Procedure: (1) whole-body FBD for reactions (2) cut at the section (3) FBD of one segment showing N, V, M with the sign convention (4) apply the three equilibrium equations

Same method-of-sections approach applies to each multi-force

member of a frame/machine – cut the individual member, not just the whole structure, to expose its own N, V, M

$\sigma = \frac{N}{A}$ – internal axial force N produces an average normal stress over area A (bridge to Mechanics of Materials)

Quick Reference – statics

Equilibrium: 2D: $\sum F_x = \sum F_y = \sum M_O = 0$ (3 eqns) – 3D: $\sum F_x = \sum F_y = \sum F_z = \sum M_x = \sum M_y = \sum M_z = 0$ (6 eqns)

Support unknowns: roller/rocker 1 – pin/hinge 2 – fixed (2D) 3 – cable/link 1 – ball-and-socket 3 – journal bearing 4 – thrust bearing 5 – fixed (3D) 6

Centroids, from the named edge/axis: rectangle $b/2, h/2$ – triangle $h/3$ – circle 0 – semicircular area $4r/3\pi$ – quarter-circular area $4r/3\pi$ (both axes) – semicircular arc $2r/\pi$ – quarter-circular arc $2r/\pi$ (both axes)

Trusses & friction: $m + r = 2j$ determinate $F \leq \mu_s N$ (static)
 $F = \mu_k N$ (kinetic) $\phi_s = \tan^{-1} \mu_s$

AER320 Statics & Strength of Materials — Final sheet, pages 1–2 of 4 (Statics, identical to the midterm sheet).

Stress, Strain & Material Behavior

$\sigma = \frac{P}{A}$ – average normal stress, axial force $P \perp$ to area A ;
 $\tau_{avg} = \frac{V}{A}$ – average shear stress, V parallel (in-plane) to A
 $\varepsilon = \frac{\delta}{L} = \frac{\Delta L}{L}$ – normal strain (elongation per unit length);
 $\gamma = \frac{\pi}{2} - \theta' - \theta$ – shear strain, change in angle between originally $\perp$ lines (rad)

$\varepsilon_T = \alpha \Delta T$ – thermal strain; α = coefficient of thermal expansion ($1/^\circ\text{C}$). Causes no stress unless deformation is restrained

Stress-strain diagram: proportional limit (Hooke's Law valid) $\rightarrow$ elastic limit $\rightarrow$ yield $\sigma_Y \rightarrow$ ultimate σ_u (max) $\rightarrow$ fracture. Ductile: yielding & necking; brittle: fractures near the elastic limit, little plastic deformation

$\sigma = E\varepsilon$, $\tau = G\gamma$ – Hooke's Law (below the proportional limit);
 E = Young's modulus, G = shear modulus

$\nu = -\frac{\varepsilon_{lat}}{\varepsilon_{long}}$ – Poisson's ratio (≈ 0.25 – 0.35 for metals); $G = \frac{E}{2(1+\nu)}$ – only 2 of E, G, ν independent (isotropic)

$$\varepsilon_x = \frac{1}{E} [\sigma_x - \nu(\sigma_y + \sigma_z)], \quad \varepsilon_y = \frac{1}{E} [\sigma_y - \nu(\sigma_x + \sigma_z)]$$

$$\varepsilon_z = \frac{1}{E} [\sigma_z - \nu(\sigma_x + \sigma_y)], \quad \gamma_{xy} = \frac{\tau_{xy}}{G}$$

– generalized Hooke's Law, isotropic (3D); also $\gamma_{yz} = \frac{\tau_{yz}}{G}$, $\gamma_{xz} = \frac{\tau_{xz}}{G}$

$\varepsilon_x = \frac{1}{E}(\sigma_x - \nu\sigma_y)$, $\varepsilon_y = \frac{1}{E}(\sigma_y - \nu\sigma_x)$, $\gamma_{xy} = \frac{\tau_{xy}}{G}$ – generalized Hooke's Law, plane stress ($\sigma_z = \tau_{xz} = \tau_{yz} = 0$)

$\varepsilon_z = -\frac{\nu}{E}(\sigma_x + \sigma_y)$ – plane stress: ε_z is generally NOT zero even though $\sigma_z = 0$

St. Venant's principle: stress concentrations near a load application point or cross-section change die out; the distribution becomes uniform at $\approx$ one cross-sectional dimension away from the discontinuity

Typical elastic constants: $E \approx 200$ GPa, $G \approx 75$ GPa, $\nu \approx 0.32$ – structural steel – $E \approx 70$ GPa, $G \approx 26$ GPa, $\nu \approx 0.35$ – aluminum alloy – $E \approx 114$ GPa, $G \approx 44$ GPa, $\nu \approx 0.31$ – titanium alloy (Ti-6Al-4V), common in aerospace – $E \approx 25$ – 30 GPa, $\nu \approx 0.15$ – 0.2 – concrete (compression)

Axial Deformation

$\delta = \frac{PL}{AE}$ – elongation, axially loaded prismatic member; constant P, A, E over length L

$$\delta = \sum_{i=1}^n \frac{N_i L_i}{A_i E_i}$$

– elongation, multiple segments: sum where N, A , or material is piecewise constant (stepped shafts, composite bars)

$$\delta = \int_0^L \frac{N(x) dx}{A(x) E}$$

– elongation, varying $N(x)$ or $A(x)$: tapered bar, bar under self-weight

$\delta_T = \alpha \Delta T L$ – free thermal expansion/contraction, unrestrained member; combine with $\delta = PL/AE$ by superposition for restrained (indeterminate) members

$\sum F = 0$ (equilibrium), $\delta_{compat} = 0$ or a specified misfit – indeterminate axial members: write equilibrium (fewer eqns than unknowns), then compatibility relating displacements; substitute $\delta = NL/AE$ ($\pm \delta_T$) and solve simultaneously

Misfits/pre-strains: a member fabricated Δ too long/short induces an internal force with zero external load; solve $\delta_{actual} = \delta_{load} \pm \Delta$ by the same compatibility approach

Equilibrium of Beams – Shear & Moment

$\frac{dV}{dx} = -w(x)$, $\frac{dM}{dx} = V(x)$ – load-shear-moment differential relations; $w(x)$ positive downward. Shear slope = –load intensity; moment slope = shear

$V_2 - V_1 = -\int_{x_1}^{x_2} w(x) dx$ – change in shear = –area under the load diagram

$M_2 - M_1 = \int_{x_1}^{x_2} V(x) dx$ – change in moment = area under the shear diagram (fast sketch of V, M without full integration)

Sign convention: positive V tends to rotate a beam element clockwise (up on the left face, down on the right); positive M bends the beam concave up (sagging), tension on the bottom fiber

Procedure: (1) find all support reactions (2) divide the beam into regions between loads/reactions/discontinuities (3) in each region cut a section and solve $V(x), M(x)$ from equilibrium (4) plot V, M vs. x , check slopes against $dV/dx, dM/dx$

$$\langle x-a \rangle^n = \begin{cases} 0, & x < a \\ (x-a)^n, & x \geq a \end{cases} \quad (n \geq 0)$$

– discontinuity (Macaulay) function; brackets "switch on" at $x = a$. Negative-exponent forms ($n = -1, -2$) represent point loads/moments and integrate by raising the exponent by one

Write the entire load $w(x)$ as one expression using Macaulay terms for each load/moment/reaction; integrate once for $V(x) = \int w dx$, again for $M(x) = \int V dx$, term by term – avoids splitting the beam into regions

Common beam cases (quick-check): $R = P/2$ each end, $M_{max} = PL/4$ at midspan – simply supported, central point load P – $R = wL/2$ each end, $M_{max} = wL^2/8$ at midspan – simply supported, uniform load w – $R = P$, $M_{max} = PL$ at the fixed end – cantilever, end point load P – $R = wL$, $M_{max} = wL^2/2$ at the fixed end – cantilever, uniform load w

Stress in Beams – Flexure

$\sigma = -\frac{My}{I}$, $\sigma_{max} = \frac{Mc}{I}$ – flexure formula; y = distance from the neutral axis (through the centroid), c = distance to the extreme fiber. Linear-elastic, symmetric bending

Assumptions: plane sections remain plane; material is linear-elastic (Hooke's Law); neutral axis passes through the centroid; strain/stress vary linearly from zero at the neutral axis

$S = \frac{I}{c}$, $\sigma_{max} = \frac{M}{S}$ – section modulus (purely geometric);

$S_{reqd} = \frac{M_{max}}{\sigma_{allow}}$ – size a section meeting S_{reqd} for beam design

$I = \int y^2 dA$, $I = \bar{I} + Ad^2$ – second moment of area & parallel-axis theorem; $\bar{I}$ about the shape's own centroidal axis, d = perpendicular distance to the parallel axis. Essential for composite/built-up sections

$J = I_x + I_y$, $k = \sqrt{\frac{I}{A}}$ – polar moment of inertia (about the same point, $\perp$ to the x - y plane) & radius of gyration

Second moment of area – common shapes: $\bar{I} = bh^3/12$ – rectangle ($b \times h$), centroidal axis $\parallel$ to base b – $I = bh^3/3$ – rectangle, about the base edge – $\bar{I} = bh^3/36$ – triangle (base b , height h), centroidal axis $\parallel$ base – $I = bh^3/12$ – triangle, about the base edge – $\bar{I} = \pi r^4/4$ – circle (radius r), any centroidal diameter – $J = \pi r^4/2$ – circle, polar, about the center – $I = \pi r^4/8$ – semicircle, about the diametral (straight) edge – $\bar{I} \approx 0.1098 r^4$ – semicircle, centroidal axis $\parallel$ diameter – $I = \pi r^4/16$ – quarter circle, about either straight edge – $\bar{I} \approx 0.0549 r^4$ – quarter circle, centroidal axis $\parallel$ a straight edge

$n = \frac{E_2}{E_1}$, $b' = n b$ – composite/non-homogeneous beams, transformed section: scale material 2's width by n to an equivalent width of material 1, then apply the ordinary flexure formula; multiply the transformed-region stress by n to recover the actual stress there

Quick Reference – strength of materials

Elastic constants: $\sigma = E\varepsilon$ – $\tau = G\gamma$ – $G = \frac{E}{2(1+\nu)}$ – $\varepsilon_T = \alpha \Delta T$

Axial & beams: $\delta = \frac{PL}{AE}$ – $\sigma_{max} = \frac{Mc}{I}$ – $S = \frac{I}{c}$ – $\frac{dV}{dx} = -w(x)$ – $\frac{dM}{dx} = V(x)$

Second moment coefficients, centroidal axis: rectangle $bh^3/12$ – triangle $bh^3/36$ – circle $\pi r^4/4$ – semicircle $\approx 0.1098 r^4$ – quarter circle $\approx 0.0549 r^4$

Beam quick-check (M_{max}): simply supported, point load $PL/4$ – simply supported, UDL $wL^2/8$ – cantilever, point load PL – cantilever, UDL $wL^2/2$

AER320 Statics & Strength of Materials — Final sheet, pages 3–4 of 4 (Strength of Materials). Comprehensive, Tests 1–4. Condensed from the full verified reference in the Formula_Ref app.